

XINHANG LU

CONTACT INFORMATION

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EDUCATION

Ph.D. in Mathematical Sciences, Nanyang Technological University, Singapore September 2021
• Thesis: *Fair Resource Allocation in Rich Domains*
• Supervisor: Xiaohui Bei

B.Eng. in Computer Science and Technology, Southeast University, Nanjing, China June 2017

APPOINTMENTS

**Department of Informatics, Faculty of Information Science and
Electrical Engineering, Kyushu University** Fukuoka, Japan
Assistant Professor, affiliated with the *Multi-Agent Laboratory* April 2026 – Present

School of Computer Science and Engineering, University of New South Wales Sydney, Australia
Postdoctoral Fellow, member of the *Algorithmic Decision Theory* and the *Algorithms* groups 2021 – 2026

Department of Computer Science, National University of Singapore Singapore
Research Fellow, hosted by Warut Suksompong 2021

AWARDS AND HONOURS

- **AAAI-20 Outstanding Student Paper Award** 2020
One paper received this award (of 4 such awards) out of 7737 submissions and 1591 accepted papers.
- **NTU Research Scholarship**, Nanyang Technological University 2017 – 2021
- **Zhang Zhiwei Scholarship**, Southeast University 2016
- **Guosheng Scholarship**, Southeast University 2015

CONFERENCE PROCEEDINGS

(α - β): Alphabetical order

- C1. **Fair Division with Indivisible Goods, Chores, and Cake.**
(α - β) Haris Aziz, Xinhang Lu, Simon Mackenzie, and Mashbat Suzuki.
In *Proceedings of the 27th ACM Conference on Economics and Computation (EC)*, July 2026. Forthcoming. Preprint available at <https://arxiv.org/abs/2511.04891>
- C2. **Approximately Fair and Population Consistent Budget Division via Simple Payment Schemes.**
(α - β) Haris Aziz, Patrick Lederer, Xinhang Lu, Mashbat Suzuki, and Jeremy Vollen.
In *Proceedings of the 26th ACM Conference on Economics and Computation (EC)*, page 349, July 2025. doi:[10.1145/3736252.3742544](https://doi.org/10.1145/3736252.3742544). The paper was accepted to the conference as a full paper but published as an abstract. Journal version in *Games and Economic Behavior (GEB)* (J2)
- C3. **Fair Allocation of Divisible Goods under Non-Linear Valuations.**
(α - β) Haris Aziz, Zixu He, Xinhang Lu, and Kaiyang Zhou.
In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 170–178, May 2025. URL <https://dl.acm.org/doi/10.5555/3709347.3743529>
- C4. **Best-of-Both-Worlds Fair Allocation of Indivisible and Mixed Goods.**
(α - β) Xiaolin Bu, Zihao Li, Shengxin Liu, Xinhang Lu, and Biaoshuai Tao.
In *Proceedings of the 20th Conference on Web and Internet Economics (WINE)*, pages 277–294, December 2024. doi:[10.1007/978-3-032-08560-3_16](https://doi.org/10.1007/978-3-032-08560-3_16)
- C5. **Welfare Loss in Connected Resource Allocation.**
(α - β) Xiaohui Bei, Alexander Lam, Xinhang Lu, and Warut Suksompong.
In *Proceedings of the 33rd International Joint Conference on Artificial Intelligence (IJCAI)*, pages 2660–2668, August 2024. doi:[10.24963/ijcai.2024/294](https://doi.org/10.24963/ijcai.2024/294). Journal version in *Discrete Applied Mathematics (DAM)* (J1)
- C6. **A Complete Landscape for the Price of Envy-Freeness.**
(α - β) Zihao Li, Shengxin Liu, Xinhang Lu, Biaoshuai Tao, and Yichen Tao.
In *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 1183–1191, May 2024. URL <https://dl.acm.org/doi/10.5555/3635637.3662975>

- C7. Fair Lotteries for Participatory Budgeting.**
 $(\alpha\text{-}\beta)$ Haris Aziz, Xinhang Lu, Mashbat Suzuki, Jeremy Vollen, and Toby Walsh.
 In *Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI)*, pages 9469–9476, February 2024. doi:[10.1609/aaai.v38i9.28801](https://doi.org/10.1609/aaai.v38i9.28801)
- C8. Mixed Fair Division: A Survey.**
 $(\alpha\text{-}\beta)$ Shengxin Liu, Xinhang Lu, Mashbat Suzuki, and Toby Walsh.
 In *Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI)*, pages 22641–22649, February 2024. doi:[10.1609/aaai.v38i20.30274](https://doi.org/10.1609/aaai.v38i20.30274). Senior Member Presentation Track. Journal version in *Journal of Artificial Intelligence Research (JAIR)* (J5)
- C9. Best-of-Both-Worlds Fairness in Committee Voting.**
 $(\alpha\text{-}\beta)$ Haris Aziz, Xinhang Lu, Mashbat Suzuki, Jeremy Vollen, and Toby Walsh.
 In *Proceedings of the 19th Conference on Web and Internet Economics (WINE)*, page 676, December 2023. [The paper was accepted to the conference as a full paper but published as an abstract.](#)
- C10. Fair Division with Subjective Divisibility.**
 $(\alpha\text{-}\beta)$ Xiaohui Bei, Shengxin Liu, and Xinhang Lu.
 In *Proceedings of the 19th Conference on Web and Internet Economics (WINE)*, page 677, December 2023. [The paper was accepted to the conference as a full paper but published as an abstract.](#) Journal version in *Games and Economic Behavior (GEB)* (J3)
- C11. Truthful Fair Mechanisms for Allocating Mixed Divisible and Indivisible Goods.**
 $(\alpha\text{-}\beta)$ Zihao Li, Shengxin Liu, Xinhang Lu, and Biaoshuai Tao.
 In *Proceedings of the 32nd International Joint Conference on Artificial Intelligence (IJCAI)*, pages 2808–2816, August 2023. doi:[10.24963/ijcai.2023/313](https://doi.org/10.24963/ijcai.2023/313)
- C12. Approval-Based Voting with Mixed Goods.**
 Xinhang Lu, Jannik Peters, Haris Aziz, Xiaohui Bei, and Warut Suksompong.
 In *Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI)*, pages 5781–5788, February 2023. doi:[10.1609/aaai.v37i5.25717](https://doi.org/10.1609/aaai.v37i5.25717). Journal version in *Social Choice and Welfare (SCW)* (J6)
- C13. Truthful Cake Sharing.**
 $(\alpha\text{-}\beta)$ Xiaohui Bei, Xinhang Lu, and Warut Suksompong.
 In *Proceedings of the 36th AAAI Conference on Artificial Intelligence (AAAI)*, pages 4809–4817, February–March 2022. doi:[10.1609/aaai.v36i5.20408](https://doi.org/10.1609/aaai.v36i5.20408). Journal version in *Social Choice and Welfare (SCW)* (J4)
- C14. The Price of Connectivity in Fair Division.**
 $(\alpha\text{-}\beta)$ Xiaohui Bei, Ayumi Igarashi, Xinhang Lu, and Warut Suksompong.
 In *Proceedings of the 35th AAAI Conference on Artificial Intelligence (AAAI)*, pages 5151–5158, February 2021. doi:[10.1609/aaai.v35i6.16651](https://doi.org/10.1609/aaai.v35i6.16651). Journal version in *SIAM Journal on Discrete Mathematics (SIDMA)* (J7)
- C15. Maximin Fairness with Mixed Divisible and Indivisible Goods.**
 $(\alpha\text{-}\beta)$ Xiaohui Bei, Shengxin Liu, Xinhang Lu, and Hongao Wang.
 In *Proceedings of the 35th AAAI Conference on Artificial Intelligence (AAAI)*, pages 5167–5175, February 2021. doi:[10.1609/aaai.v35i6.16653](https://doi.org/10.1609/aaai.v35i6.16653). Journal version in *Autonomous Agents and Multi-Agent Systems (JAAMAS)* (J10)
- C16. Fair Division of Mixed Divisible and Indivisible Goods.**
 $(\alpha\text{-}\beta)$ Xiaohui Bei, Zihao Li, Jinyan Liu, Shengxin Liu, and Xinhang Lu.
 In *Proceedings of the 34th AAAI Conference on Artificial Intelligence (AAAI)*, pages 1814–1821, February 2020. doi:[10.1609/aaai.v34i02.5548](https://doi.org/10.1609/aaai.v34i02.5548). Invited for publication in *Artificial Intelligence (AIJ)* through the fast track scheme (J11)
[AAAI-20 Outstanding Student Paper Award](#)
- C17. The Price of Fairness for Indivisible Goods.**
 $(\alpha\text{-}\beta)$ Xiaohui Bei, Xinhang Lu, Pasin Manurangsi, and Warut Suksompong.
 In *Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 81–87, August 2019. doi:[10.24963/ijcai.2019/12](https://doi.org/10.24963/ijcai.2019/12). Journal version in *Theory of Computing Systems (TOCS)* (J9)

- J1. **Welfare Loss in Connected Resource Allocation.**
(α - β) Xiaohui Bei, Alexander Lam, Xinhang Lu, and Warut Suksompong.
Discrete Applied Mathematics (DAM), 385:1–23, May 2026. doi:[10.1016/j.dam.2026.01.007](https://doi.org/10.1016/j.dam.2026.01.007). Preliminary version in IJCAI-24 (C5)
- J2. **Approximately Fair and Population Consistent Budget Division via Simple Payment Schemes.**
(α - β) Haris Aziz, Patrick Lederer, Xinhang Lu, Mashbat Suzuki, and Jeremy Vollen.
Games and Economic Behavior (GEB), 154:208–225, December 2025. doi:[10.1016/j.geb.2025.09.001](https://doi.org/10.1016/j.geb.2025.09.001). Preliminary version in EC-25 (C2)
- J3. **Fair Division with Subjective Divisibility.**
(α - β) Xiaohui Bei, Shengxin Liu, and Xinhang Lu.
Games and Economic Behavior (GEB), 151:127–147, May 2025. doi:[10.1016/j.geb.2025.03.004](https://doi.org/10.1016/j.geb.2025.03.004). Preliminary version in WINE-23 (C10)
- J4. **Truthful Cake Sharing.**
(α - β) Xiaohui Bei, Xinhang Lu, and Warut Suksompong.
Social Choice and Welfare (SCW), 64(1–2):309–343, February 2025. doi:[10.1007/s00355-023-01503-0](https://doi.org/10.1007/s00355-023-01503-0). Special Issue on Fair Public Decision Making: Allocating Budgets, Seats, and Probability. Preliminary version in AAAI-22 (C13)
- J5. **Mixed Fair Division: A Survey.**
(α - β) Shengxin Liu, Xinhang Lu, Mashbat Suzuki, and Toby Walsh.
Journal of Artificial Intelligence Research (JAIR), 80:1373–1406, August 2024. doi:[10.1613/jair.1.15800](https://doi.org/10.1613/jair.1.15800). Preliminary version in AAAI-24 (C8)
- J6. **Approval-Based Voting with Mixed Goods.**
Xinhang Lu, Jannik Peters, Haris Aziz, Xiaohui Bei, and Warut Suksompong.
Social Choice and Welfare (SCW), 62(4):643–677, June 2024. doi:[10.1007/s00355-024-01511-8](https://doi.org/10.1007/s00355-024-01511-8). Preliminary version in AAAI-23 (C12)
- J7. **The Price of Connectivity in Fair Division.**
(α - β) Xiaohui Bei, Ayumi Igarashi, Xinhang Lu, and Warut Suksompong.
SIAM Journal on Discrete Mathematics (SIDMA), 36(2):1156–1186, 2022. doi:[10.1137/20M1388310](https://doi.org/10.1137/20M1388310). Preliminary version in AAAI-21 (C14)
- J8. **Throughput Maximization in Wireless Communication Systems Powered by Hybrid Energy Harvesting.**
Chenchen Fu*, Xinhang Lu*, Xiaoxing Qiu, Sujunjie Sun, Xueyong Xu, Weiwei Wu, Chun Jason Xue, and Song Han.
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 41(11):3981–3992, November 2022. doi:[10.1109/TCAD.2022.3197978](https://doi.org/10.1109/TCAD.2022.3197978). The asterisk (*) denotes equal contribution.
- J9. **The Price of Fairness for Indivisible Goods.**
(α - β) Xiaohui Bei, Xinhang Lu, Pasin Manurangsi, and Warut Suksompong.
Theory of Computing Systems (TOCS), 65(7):1069–1093, October 2021. doi:[10.1007/s00224-021-10039-8](https://doi.org/10.1007/s00224-021-10039-8). Preliminary version in IJCAI-19 (C17)
- J10. **Maximin Fairness with Mixed Divisible and Indivisible Goods.**
(α - β) Xiaohui Bei, Shengxin Liu, Xinhang Lu, and Hongao Wang.
Autonomous Agents and Multi-Agent Systems (JAAMAS), 35(2):34, October 2021. doi:[10.1007/s10458-021-09517-7](https://doi.org/10.1007/s10458-021-09517-7). Special Issue on Fair Division. Preliminary version in AAAI-21 (C15)
- J11. **Fair Division of Mixed Divisible and Indivisible Goods.**
(α - β) Xiaohui Bei, Zihao Li, Jinyan Liu, Shengxin Liu, and Xinhang Lu.
Artificial Intelligence (AIJ), 293:103436, April 2021. doi:[10.1016/j.artint.2020.103436](https://doi.org/10.1016/j.artint.2020.103436). Preliminary version in AAAI-20 (C16)
- J12. **The Anatomy of the Global Football Player Transfer Network: Club Functionalities versus Network Properties.**
Xiaofan Liu, Yuliang Liu, Xinhang Lu, Qixuan Wang, and Tongxing Wang.
PLOS ONE, 11(6):e0156504, June 2016. doi:[10.1371/journal.pone.0156504](https://doi.org/10.1371/journal.pone.0156504)

NEWSLETTER (α - β): Alphabetical order	N1. M-PREF 2023: 14th Multidisciplinary Workshop on Advances in Preference Handling – A Vivid Workshop Held in Macao, S.A.R., Between Two Former Islands. (α-β) Haris Aziz, Ulrich Junker, Xinhang Lu, Nicholas Mattei, and Andrea Passerini. <i>IFORS Newsletter</i>, 18(4):33–34, Dec. 2023. URL ifors.org/newsletter/ifors-news-dec-2023
SUPERVISION EXPERIENCES	1 UNSW undergraduate Taste of Research project co-supervised with Haris Aziz 2024 – Collaboration led to (C3). 1 UNSW Honours Thesis co-supervised with Haris Aziz 2022 – 2023 1 NUS Undergraduate Research Programme Project co-mentored with Warut Suksompong 2021
TEACHING EXPERIENCES	Project Mentor, School of Computer Science and Engineering, UNSW Sydney - COMP3821/9801: Extended Algorithm Design and Analysis Term 3, 2025 Guest Lecturer - UNSW COMP4920: Professional Issues and Ethics in Information Technology March 2024 - NUS CS 6235: Topics in Computational Social Choice February 2021 & March 2023 Teaching Assistant, Division of Mathematical Sciences, Nanyang Technological University (Awarded the University Teaching for Teaching Assistant Certificate in 2018.) - MAS 714: Algorithms and Theory of Computation Fall 2020 - MH4320: Computational Economics Fall 2019, 2020 - MH2500: Probability and Introduction to Statistics Fall 2019 - MH1812: Discrete Mathematics Fall 2019 - MH1811: Mathematics 2 Spring 2019 - MH1810: Mathematics 1 Fall 2018
SERVICE AND OUTREACH	Workshop Organization <i>14th Multidisciplinary Workshop on Advances in Preference Handling (M-PREF)</i> at IJCAI-23; see (N1). Tutorial Organization <i>Recent Developments in Mixed Fair Division</i> at WINE-23 and AAMAS-24. <i>Developments in Fair Resource Allocation</i> at AJCAI-22. Program Committee Member - AAAI Conference on Artificial Intelligence (AAAI) 2021 – 2024 - International Joint Conference on Artificial Intelligence (IJCAI) 2022 – 2024 - International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2023 - European Conference on Artificial Intelligence (ECAI) 2024 - IJCAI Workshop on Computational Fair Division 2023, 2024 - International Joint Conference on Theoretical Computer Science – Frontier of Algorithmic Wisdom (IJTCS-FAW) 2023 Journal Referee Algorithmica, Artificial Intelligence (AIJ), Autonomous Agents and Multi-Agent Systems (JAAMAS), Games and Economic Behavior (GEB), Information and Computation, Journal of Artificial Intelligence Research (JAIR), Mathematical Social Sciences Conference Reviewer AAMAS (2022), COCOA (2020), EAAMO (2022), ESA (2022), FSTTCS (2021), ICALP (2024), IPCO (2024), ISAAC (2019), MATCHUP (2022), NCTCS (2019), SAGT (2021, 2022), SODA (2021, 2026), WINE (2020, 2022)
INVITED TALKS & SELECTED PRESENTATIONS (excl. conference talks)	Fair Division with Indivisible Goods, Chores, and Cake The 18th Meeting of the Society for Social Choice and Welfare, Tokyo, Japan June 2026

Best-of-Both-Worlds Fair Allocation of Indivisible and Mixed Goods

- Algorithmics of Fair Division & Social Choice, Inst. for Mathematical Sciences, NUS November 2024
- Sydney Algorithms and Computing Theory Group, The University of Sydney October 2024

Recent Developments in Mixed Fair Division

- Tutorial Track at the 23rd AAMAS, Auckland, New Zealand May 2024
- Tutorial Track at the 19th WINE, Shanghai, China December 2023

Fair Division with Subjective Divisibility

- The 2nd IJCAI Workshop on Computational Fair Division, Jeju, South Korea August 2024
- Inst. for Theoretical Computer Science, Shanghai Uni. of Finance and Economics November 2023

Fair Division of Mixed Goods: Justice, Truth, and Beyond

- Multi-Agent Laboratory, Graduate School of ISEE, Kyushu University September 2025
- Sydney Algorithms and Computing Theory Group, The University of Sydney October 2023

Truthful Fair Mechanism for Allocating Mixed Divisible and Indivisible Goods

- Workshop on Game Theory and Fair Division, The Hong Kong Polytechnic University May 2023

Best-of-Both-Worlds Fairness in Committee Voting

- Reading Group in the Department of Computer Science, City University of Hong Kong May 2023
- NUS CS 6235: Topics in Computational Social Choice March 2023

Developments in Fair Resource Allocation

- Tutorial Track at the 35th AJCAI, Perth, Australia December 2022

Approval-Based Voting with Mixed Goods

- Summer School on Algorithmic Game Theory at City University of Hong Kong, Virtual June 2023
- Centre for Mathematical Social Science, The University of Auckland December 2022

Truthful Cake Sharing

- Computational and Network Economics Track at IJTCS-FAW, Virtual August 2022
- QuACT Seminar in the Institute of Computing Technology at CAS, Virtual March 2022

Maximin Fairness with Mixed Divisible and Indivisible Goods

- Young PhD Forum at IJTCS, Virtual August 2021

Fair Division of Mixed Divisible and Indivisible Goods

- Workshop on Fair Resource Allocation: Concept, Algorithms and Complexity at EC, Virtual July 2021
- NUS CS 6235: Topics in Computational Social Choice February 2021